

“2024 - 07 -20”

Four taxa and two areas

Now, we load the data included in the package: tree and distribution.

```
## Libraries
```

```
library(blepd)
```

```
## Loading required package: ape
```

```
## Loading required package: picante
```

```
## Loading required package: vegan
```

```
## Loading required package: permute
```

```
## Loading required package: lattice
```

```
## This is vegan 2.6-6.1
```

```
## Loading required package: nlme
```

```
library(ggtree)
```

```
## ggtree v3.10.1 For help: https://yulab-smu.top/treedata-book/
```

```
##
```

```
## If you use the ggtree package suite in published research, please cite  
## the appropriate paper(s):
```

```
##
```

```
## Guangchuang Yu, David Smith, Huachen Zhu, Yi Guan, Tommy Tsan-Yuk Lam.
```

```
## ggtree: an R package for visualization and annotation of phylogenetic
```

```
## trees with their covariates and other associated data. Methods in
```

```
## Ecology and Evolution. 2017, 8(1):28-36. doi:10.1111/2041-210X.12628
```

```
##
```

```
## Guangchuang Yu, Tommy Tsan-Yuk Lam, Huachen Zhu, Yi Guan. Two methods
```

```
## for mapping and visualizing associated data on phylogeny using ggtree.
```

```
## Molecular Biology and Evolution. 2018, 35(12):3041-3043.
```

```
## doi:10.1093/molbev/msy194
```

```
##
```

```
## S Xu, Z Dai, P Guo, X Fu, S Liu, L Zhou, W Tang, T Feng, M Chen, L
```

```
## Zhan, T Wu, E Hu, Y Jiang, X Bo, G Yu. ggtreeExtra: Compact
```

```
## visualization of richly annotated phylogenetic data. Molecular Biology
```

```
## and Evolution. 2021, 38(9):4039-4042. doi: 10.1093/molbev/msab166
```

```
##
```

```
## Attaching package: 'ggtree'
```

```
## The following object is masked from 'package:nlme':
```

```
##
```

```
## collapse
```

```
## The following object is masked from 'package:ape':
```

```
##
```

```
## rotate
```

```
library(ggplot2)
```

```
## Data

# Call the 4 terminals example

data(package = "blepd")

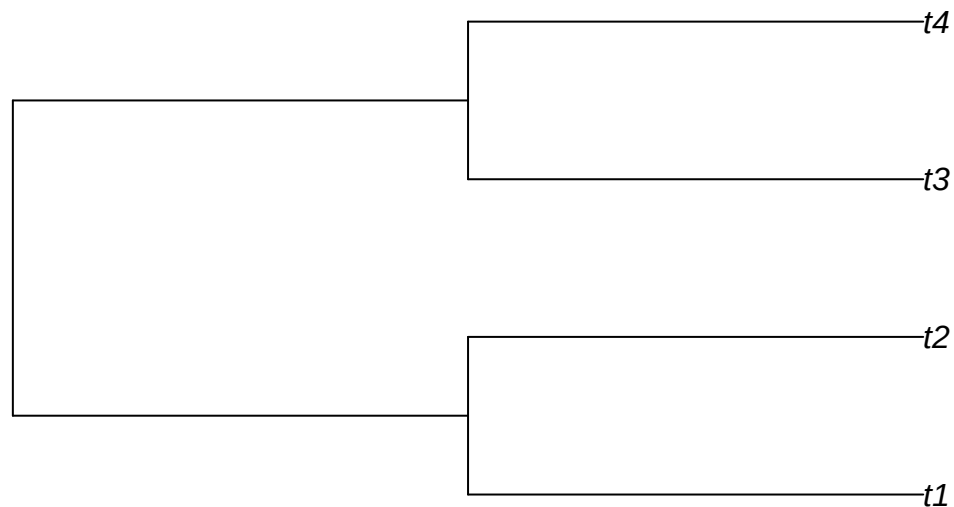
## trees

data(tree)

str(tree)

## List of 5
## $ edge      : int [1:6, 1:2] 5 6 6 5 7 7 6 1 2 7 ...
## $ edge.length: num [1:6] 1 1 1 1 1 1
## $ Nnode      : int 3
## $ tip.label  : chr [1:4] "t1" "t2" "t3" "t4"
## $ root.edge  : num 1
## - attr(*, "class")= chr "phylo"
## - attr(*, "order")= chr "cladewise"

plot(tree,use.edge.length =T)
```



```
initialTree <- tree
```

```

## distributions

data(distribution)

str(distribution)

##  int [1:2, 1:4] 1 0 0 1 1 0 0 1
##  - attr(*, "dimnames")=List of 2
##    ..$ : chr [1:2] "A1" "A2"
##    ..$ : chr [1:4] "t1" "t2" "t3" "t4"

dist4taxa <- distribution

## distribution to XY to be able to plot the data, otherwise, skip the next step

distXY <- matrix2XY(dist4taxa)

distXY

##      Terminal Area
## 1      t1      A1
## 2      t3      A1
## 3      t2      A2
## 4      t4      A2

## plotting

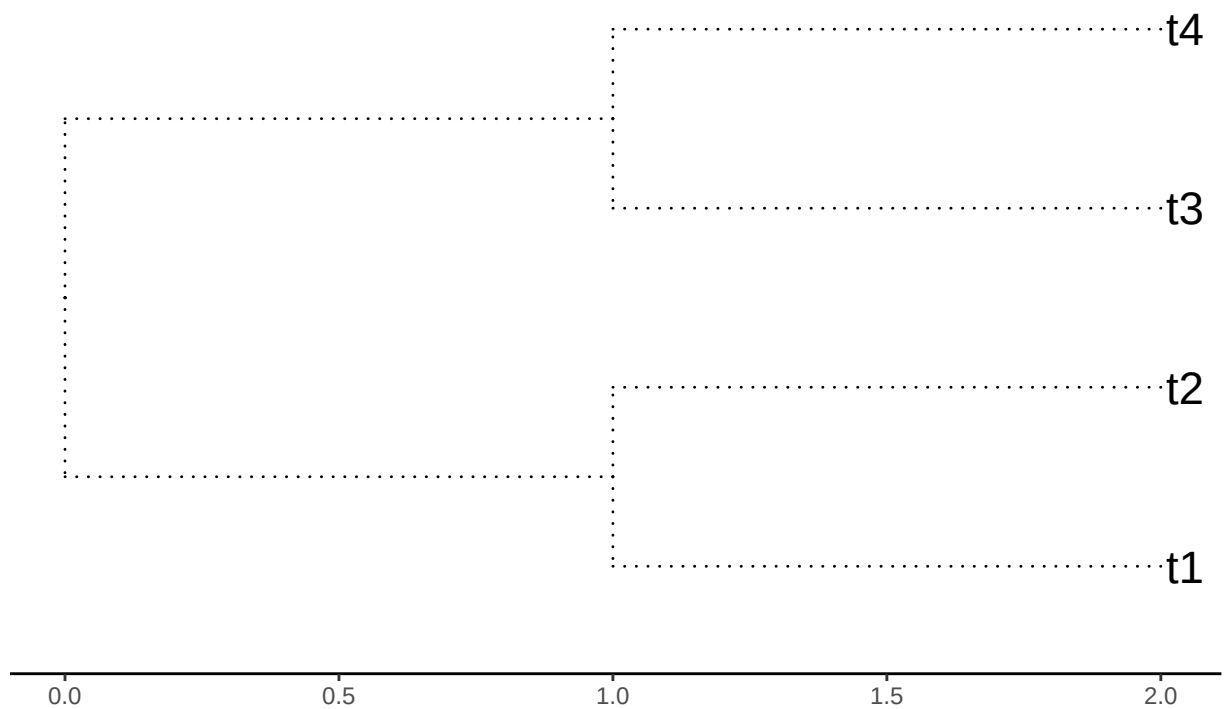
## the tree

plotTree <- ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=6, color="black") +
  theme_tree2() +
  ggtitle("Four terminals, equal branch length")

##
print(plotTree)

```

Four terminals, equal branch length

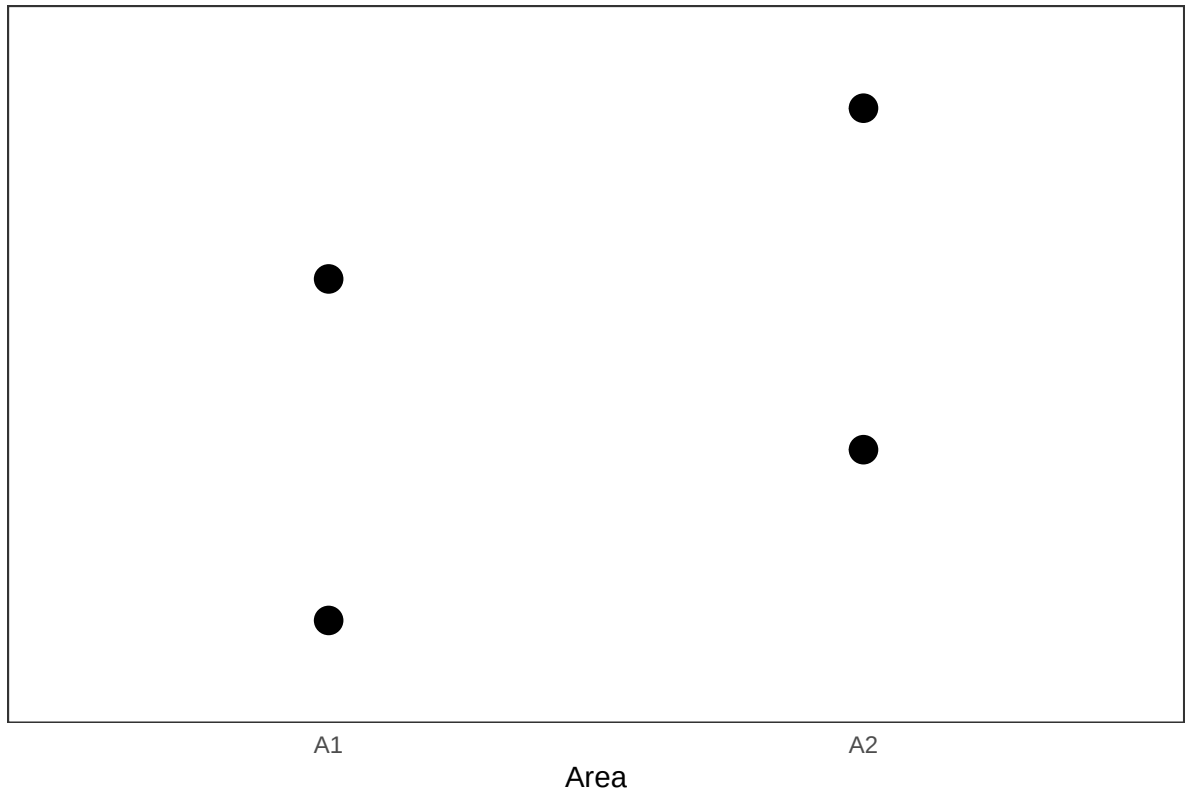


```
## the distribution

plotDistrib <- ggplot(data=distXY,
                      aes(x= Area, y= Terminal , size =150)) +
  geom_point() +
  theme_bw() +
  theme( axis.text.y = element_blank(),
        panel.grid.major = element_blank(), # Remove major gridlines
        panel.grid.minor = element_blank(), # Remove minor gridlines
        axis.line = element_line(colour = NA_character_), # Remove axis lines
        axis.ticks = element_blank(), # Remove axis ticks
        legend.position = "none"
  ) +
  labs(title = "Distributions",
       y = "",
       x = "Area")

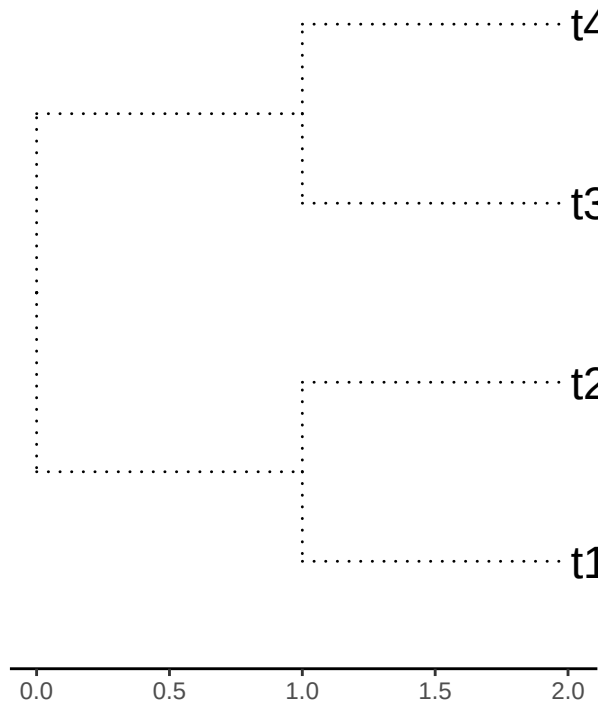
##
print(plotDistrib)
```

Distributions

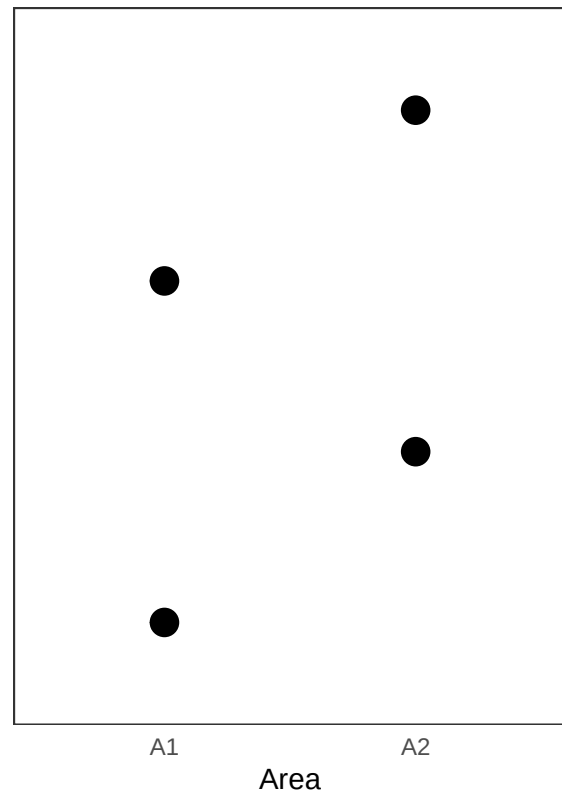


```
cowplot::plot_grid(plotTree, plotDistrib, ncol=2)
```

Four terminals, equal branch length



Distributions



We check whether names in both objects: initialTree and dist4taxa, are the same.

```
##all(colnames(dist4taxa) == initialTree$tip.label)
all(colnames(dist4taxa) %in% initialTree$tip.label)
```

```
## [1] TRUE
```

We report the branch length, and calculate the PD values.

```
initialTree$edge.length
```

```
## [1] 1 1 1 1 1 1
```

```
initialPD <- PDindex(tree=initialTree, distribution = dist4taxa)
```

```
initialPD
```

```
## [1] 4 4
```

As expected there is a tie between both areas.

Now, we can see the impact of the possible null model(s), we will use the three available models to swap: “simpleswap”, “allswap”, “uniform”, and we will swap the three classes of branches: “terminals”, “internals”, “all”.

?swapBL

```
for( modelo in c("simpleswap", "allswap", "uniform") ){
  for( rama in c("terminals", "internals", "all") ){
```

```

        val <- swapBL(tree=initialTree,
                      distribution = dist4taxa,
                      model = modelo,
                      branch = rama
                      )

        cat( "\n\t Model=\t\t",modelo,
              "\n\t Branchs swapped=\t",rama,"\n" )

        printBlepd(val)

        cat( "\n\n\n" )

    }

}

```

```

## model to test simpleswap reps 100
##
##   Model=      simpleswap
##   Branchs swapped=  terminals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
## model to test simpleswap reps 100
##
##   Model=      simpleswap
##   Branchs swapped=  internals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
## model to test simpleswap reps 100
##
##   Model=      simpleswap
##   Branchs swapped=  all
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
## model to test allswap reps 100
##
##   Model=      allswap
##   Branchs swapped=  terminals

```

```

##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
## model to test allswap reps 100
##
##   Model=      allswap
##   Branchs swapped=  internals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
## model to test allswap reps 100
##
##   Model=      allswap
##   Branchs swapped=  all
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
## model to test uniform reps 100
##
##   Model=      uniform
##   Branchs swapped=  terminals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
## model to test uniform reps 100
##
##   Model=      uniform
##   Branchs swapped=  internals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
## model to test uniform reps 100
##
##   Model=      uniform

```



```
## Branchs swapped=    all
##
## BestInitial:A1A2
## AreaSelected Freq Percent
## 1          A1A2 100      100
```

As expected, as all braches are EQUAL nothing happens

Now, let us see if the results could be assigned to the longest branches (or not).

```
## this is?
## source("/home/rafael/disco2/proyectosDRME/indices/blepd/docs/print00.r")
```

```
## we could use a single command (*evalBranch*)
```

```
?evalBranch
```

```
AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")
```

```
## Loading required package: maps
```

```
##
```

```
## Attaching package: 'phytools'
```

```
## The following object is masked from 'package:vegan':
```

```
##
```

```
## scores
```

```
str(AllTerminals)
```

```
## List of 2
## $ lower:List of 4
## ..$ : 'data.frame': 1 obs. of 5 variables:
## .. ..$ branchToEval: chr "[t1]|1"
## .. ..$ resp : chr "A1A2"
## .. ..$ resp.1 : chr "A2"
## .. ..$ approach : chr "lower"
## .. ..$ delta : num -1.19
## ..$ : 'data.frame': 1 obs. of 5 variables:
## .. ..$ branchToEval: chr "[t2]|2"
## .. ..$ resp : chr "A1A2"
## .. ..$ resp.1 : chr "A1"
## .. ..$ approach : chr "lower"
## .. ..$ delta : num -1.19
## ..$ : 'data.frame': 1 obs. of 5 variables:
## .. ..$ branchToEval: chr "[t3]|3"
## .. ..$ resp : chr "A1A2"
## .. ..$ resp.1 : chr "A2"
## .. ..$ approach : chr "lower"
## .. ..$ delta : num -1.19
## ..$ : 'data.frame': 1 obs. of 5 variables:
## .. ..$ branchToEval: chr "[t4]|4"
## .. ..$ resp : chr "A1A2"
```

```

## .. ..$ resp.1      : chr "A1"
## .. ..$ approach    : chr "lower"
## .. ..$ delta       : num -1.19
## $ upper:List of 4
## ..$ : 'data.frame': 1 obs. of 5 variables:
## .. ..$ branchToEval: chr "[t1]|1"
## .. ..$ resp        : chr "A1A2"
## .. ..$ resp.1      : chr "A1"
## .. ..$ approach    : chr "upper"
## .. ..$ delta       : num 1.59
## ..$ : 'data.frame': 1 obs. of 5 variables:
## .. ..$ branchToEval: chr "[t2]|2"
## .. ..$ resp        : chr "A1A2"
## .. ..$ resp.1      : chr "A2"
## .. ..$ approach    : chr "upper"
## .. ..$ delta       : num 1.59
## ..$ : 'data.frame': 1 obs. of 5 variables:
## .. ..$ branchToEval: chr "[t3]|3"
## .. ..$ resp        : chr "A1A2"
## .. ..$ resp.1      : chr "A1"
## .. ..$ approach    : chr "upper"
## .. ..$ delta       : num 1.59
## ..$ : 'data.frame': 1 obs. of 5 variables:
## .. ..$ branchToEval: chr "[t4]|4"
## .. ..$ resp        : chr "A1A2"
## .. ..$ resp.1      : chr "A2"
## .. ..$ approach    : chr "upper"
## .. ..$ delta       : num 1.59

#### so far so good

## this part below is strange

#####

printEvalBranch(AllTerminals)

##      node initialArea FinalArea Aproach %Delta
## 1 [t1]|1      A1A2      A2  lower  -1.19
## 2 [t2]|2      A1A2      A1  lower  -1.19
## 3 [t3]|3      A1A2      A2  lower  -1.19
## 4 [t4]|4      A1A2      A1  lower  -1.19
## 5 [t1]|1      A1A2      A1  upper   1.59
## 6 [t2]|2      A1A2      A2  upper   1.59
## 7 [t3]|3      A1A2      A1  upper   1.59
## 8 [t4]|4      A1A2      A2  upper   1.59

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)

```

##	node	initialArea	FinalArea	Approach	%Delta
## 1	[node number:6:t1/t2] 6	A1A2	A1A2	lower	0
## 2	[node number:7:t3/t4] 7	A1A2	A1A2	lower	0
## 3	[node number:6:t1/t2] 6	A1A2	A1A2	upper	0
## 4	[node number:7:t3/t4] 7	A1A2	A1A2	upper	0

```
AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="all",
                           approach="all")
```

```
printEvalBranch(AllInternals)
```

##	node	initialArea	FinalArea	Approach	%Delta
## 2	[t1] 1	A1A2	A2	lower	-1.19
## 3	[t2] 2	A1A2	A1	lower	-1.19
## 5	[t3] 3	A1A2	A2	lower	-1.19
## 6	[t4] 4	A1A2	A1	lower	-1.19
## 8	[t1] 1	A1A2	A1	upper	1.59
## 9	[t2] 2	A1A2	A2	upper	1.59
## 11	[t3] 3	A1A2	A1	upper	1.59
## 12	[t4] 4	A1A2	A2	upper	1.59

This is the same